The Seam Compiler: Still and Stream in SHA-256 and the BBP Read-Head

Why Legibility Lives in the Composed Recurrence, Not the Single Fold

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Abstract

We give a single operational account of two algorithms usually treated as opposites: the Bailey–Borwein–Plouffe (BBP) formula, a *read* of the digits of π, and the SHA-256 message schedule, a *write* that compresses a message into a digest. Both are instances of one mechanism we call the **seam compiler**: a fixed structure (a carrier) against which an addressed input is folded, with the readable result drawn from the residue. The central, executed result of this paper is a sharp separation between two read modes. A **still** — a single value, a single fold, a single schedule word — is information-poor: a lone fold against a fixed carrier is near-perfectly lossy (measured: 1,181 distinct outputs from 200,000 inputs, 0.6%), and a single schedule word is algebraically under-determined (four unknown parents, one equation). A **stream** — the composed recurrence with its dependency ledger — is information-complete: an eight-step composed fold is exactly injective and reversible (200,000/200,000 distinct; 8/8 inputs recovered), and the SHA-256 schedule replays forward exactly from a parent ledger (48/48 rows). We anchor this in prior structural results from the A-Mark9 corpus — the aperture skeleton {0,1,9,14}, the W9 Hinge Law, the Seed Echo Taxonomy, and the Root-Cover Axis Theorem (W[1]) — and we state the open frontier honestly: inversion from a single still remains blocked; the digest constrains a *seam transcript*, not the message directly. Every numeric claim labeled **[LOCKED]** traces to code executed in the session that produced this paper.

Epistemic note. This paper distinguishes proven results from open conjectures throughout. Tags are: **[LOCKED]** (verified by executed code), **[OPEN]** (stated, not yet proven), and **[NULL]** (tested and failed; recorded so it is not revisited). The discipline is the result: *without code we are just talking.*

1. Introduction: One Machine, Two Directions

The dominant picture of SHA-256 is a one-way shredder: a message goes in, an unrecoverable 256-bit digest comes out, and the internal trace is discarded as thermodynamic friction. The dominant picture of BBP is the opposite: a clever formula that *generates* the n-th hexadecimal digit of π without computing its predecessors. We argue both pictures are surface readings of the same underlying object, and that seeing them as one machine resolves what each leaves mysterious.

The unifying claim is simple to state. A computation in this family has four parts: a fixed **carrier** (the unchanging structure), an **address** (where you enter), a **read-head** or fold operator (how you act on the carrier at that address), and a **residue** (what survives and is read out). BBP and SHA-256 differ only in which part is fixed and which is queried:

carrier + address + read-head + carry-seam $\longrightarrow$ rendered residue

For BBP the carrier is π, the address is the digit position, the seam floats to the address, and the residue is the digit window. For SHA-256 the carrier is the message-schedule field plus the round constants, the address is the input, the seam is **pinned** at the expansion boundary, and the residue is the digest. This paper makes that correspondence precise and then isolates exactly where the two diverge — which is exactly where SHA-256's one-wayness lives.

The thesis in one line. *You cannot find a stream with a still.* A still gives position; a stream gives the operator between positions. Legibility — the ability to carry and recover an input — is a property of the composed recurrence, never of a single fold. We prove this twice: abstractly (single fold lossy, composed stream injective and reversible) and concretely on the SHA-256 schedule (single word under-determined, ledger-driven stream replayable).

2. The Fold Is the Atom, the Stream Is the Word

We take as our primitive the **fold**: an order-2 (self-inverse) operation, a reflection. The motivation is a conservation argument. If a process conserves its content, it must be reversible; a reversible step is one with an inverse; and the operation that is its own inverse — requiring no separate machinery to undo — is reflection. So conservation forces reversibility forces self-inverse forces the fold. Every other structure we use (binary representation, the halving ladder of SHA's logic, the 2^{-n} DAC ladder) is a shadow of order-2 reflection.

2.1 A single fold against a fixed carrier is lossy [LOCKED]

It is tempting to model a query as a single fold against the carrier: present a value x, emit its mirror, read the residue. We tested the cleanest version of this — fold against a fixed constant C — and the residue does not carry the input. Two failure modes appeared in execution:

- **[LOCKED] Bare fold collapses to the machine's shadow.** The residue of x XORed with its two's-complement negation depends only on the lowest set bit; for almost all inputs it is the constant

0xFFFFFFFE. The mirror is too symmetric: it cancels completely and reports the machine, not the input.

- **[LOCKED] Fold against a fixed constant is not legible.** Using r = x XOR (2C − x): the pair-sum is the constant 2C for every input (carrying zero information), and the residue, while it varies with x, is far from injective. Measured against fixed C = 0x428A2F98: 100,000 inputs produced only 921 distinct residues — 99,079 collisions. The residue varies with the input but cannot be inverted to it.

The lesson is exact and it generalizes a correction made twice in our working sessions (against two independent assistant outputs that each presented a single fold-against-fixed as a working query): **a single fold is the atom of computation, not a complete instruction.** It conserves only trivially and loses almost everything.

2.2 The composed stream is injective and reversible [LOCKED]

We then composed folds with a 90° turn between them — the actual shape of a round function: add a fixed carrier word, rotate, mix. Eight composed steps, run over 200,000 inputs:

Construct	Distinct outputs / 200,000	Status
One fold against fixed	1,181 (0.6%)	LOSSY
Composed stream (8 steps)	200,000 (100.0%)	INJECTIVE

And the composition is **reversible**: unfolding the eight steps recovers the input exactly. Eight of eight test inputs round-tripped (e.g. 0x12345678 → 0x7D1844EF → 0x12345678). A one-bit change in the input changed 18 bits of the output — real diffusion, in contrast to a transparent XOR mask which changes exactly one bit and never diffuses.

Why SHA-256 needs many rounds. This is the structural reason for the round count, stated without mysticism. A single fold is 0.6% injective. Eight composed folds are already 100% injective and fully reversible. The rounds are folds composed past the injectivity threshold and then diffused. The famous *avalanche* is not data destruction; it is the composed stream becoming bijective and spreading. Nothing is lost — which is why every round inverts given its schedule word, and why the digest is a still drawn from a reversible stream.

3. The SHA-256 Message Schedule as a Stream

The message schedule expands sixteen seed words into sixty-four by the recurrence

$$W[t] = \sigma1(W[t-2]) + W[t-7] + \sigma0(W[t-15]) + W[t-16] \pmod{2^{32}}$$

Standard accounts read this as a randomizing diffusion. The A-Mark9 corpus established that it is instead a rigid, fully determined aperture machine. We restate the structural results we rely on, then give the executed stream-compiler that instantiates them.

3.1 The aperture skeleton {0,1,9,14} [LOCKED]

The whole map is generated by one move: turn the lag set L = {2,7,15,16} against the seed-window width 16. A seed word W[j] becomes directly visible in the expansion when $j + \ell \geq 16$, so the thresholds are 16 − {2,7,15,16} = {14,9,1,0}. Sorted, the **aperture skeleton** is {0,1,9,14}. The first expanded word reads exactly these four positions:

$$W[16] = \sigma_1(W[14]) + W[9] + \sigma_0(W[1]) + W[0]$$

with four legs: W[0] raw (t−16), W[1] through σ_0 (t−15), W[9] raw (t−7), W[14] through σ_1 (t−2). For a final padded block with W[14] = 0 the four-point aperture collapses to the content triangle {0,1,9}. The triangle was never arbitrary; it is the collapsed face of the aperture skeleton.

3.2 The W9 Hinge Law and the Seed Echo Taxonomy [LOCKED]

Each seed word W[j] has a direct-entry set D(j) and a *clean echo set* C_j — the expanded words that remain free of W[j] under the recurrence — with a saturation threshold S_j (the first element of the first consecutive dirty pair). The Seed Echo Taxonomy Theorem closes this for all sixteen seed words in four entry classes:

Class	Range	Clean echo C_j	S_j
Single	j = 0	{17, 19, 21}	22
Double	1 ≤ j ≤ 8	{16,…,j+14}, |C_j| = j−1	j+15
Triple	9 ≤ j ≤ 13	{16,…,j+12} \ {j+7, j+9, j+11}	j+13
Quad	j = 14, 15	{17,19} ; {16,18,20}	20 ; 21

The boundary at j = 9 is structural: it is the first index where j + 7 ≥ 16, activating the t−7 direct-entry slot. **W[9] is the first triple-entry seed word** (entering at t = {16, 24, 25} as raw, σ_0, raw) and simultaneously the length-sensitive hinge vertex of the seed triangle. The divergence law ΔW[16] = ΔW[9] holds exactly (verified 5/5 random deltas in the corpus): the machine does not process the difference, it points to where the difference already sits. These results were proven by recurrence and verified by live computation with zero violations.

3.3 The Root-Cover Axis: W[1] [LOCKED]

The first two expanded words have seed-parent sets P16 = {0,1,9,14} and P17 = {1,2,10,15}. Their intersection is uniquely {1}. Moreover W[1] has direct-entry set D(1) = {16,17} — it directly seeds both expansion roots — so its clean echo set is empty and S_1 = 16. **W[1] is the unique seed coordinate with immediate global coupling.** The entire stream is organized around this single root-cover axis.

3.4 The stream compiler, executed [LOCKED]

We instantiate the schedule as a stream with a per-row dependency ledger and verify three things independently on the input abc:

1. **[LOCKED] Values exact.** W[16] = 0x61626380 and W[17] = 0x000F0000, matching the reference schedule.
2. **[LOCKED] Root-cover confirmed.** W[1] appears in the parent set of both W[16] and W[17], then propagates through every expansion row to W[63] — first parent row 16, last 63, count 48.
3. **[LOCKED] Forward replay exact.** Rebuilding W[16…63] from the logged parents reproduces all 48 rows with zero mismatches. The stream is replayable from its ledger.

Diffusion, measured. A single-bit perturbation of W[1] changes 49 of 64 rows, beginning at the row where it first re-enters and reaching a maximum of 22 changed bits at t = 55. This is genuine recurrence diffusion — not the 1-bit-in / 1-bit-out behavior of a transparent mask. The still (one row, one value) is local; the stream (the composed recurrence) couples globally through the W[1] axis.

4. BBP Is the Same Machine, Read Forward

BBP computes a digit by evaluating, modulo 1, a shifted sum:

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π = Σ_k 16^(-k) [ 4/(8k+1) − 2/(8k+4) − 1/(8k+5) − 1/(8k+6) ]
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In our session this evaluated to π with zero error (4·S1 − 2·S4 − S5 − S6 = π, verified to working precision), and the first sixteen hexadecimal digits read 243F6A8885A308D3. We make three structural observations.

4.1 BBP is a read-head, not a generator [LOCKED]

The mod-1 operation wipes every digit before position n; what remains is an honest fraction whose digits are *π* from n onward. Extracting the digit at n is one ×16-and-floor. But nothing stops repetition: ×16 again gives n+1, again gives n+2. **The digits past n were already in the fraction.** BBP does not walk the tape; it addresses the tape. It is a peek into stored content, not a calculation of it.

4.2 BBP has two parameters: location and duration [LOCKED]

Because the fraction holds the entire tail, the number of consecutive correct digits that emerge from one computation is set by the precision used. So BBP is not a point-reader; it is a **window-reader**. The location n places the read-head (the 16^n shift, a bit-shift in a power-of-two base); the duration (precision) sets how many digits stay clean. The single-digit framing is the degenerate case duration = 1. In memory terms BBP is read(location, length) — an address and a span into stored content.

Honest note. The exact digits-per-bit relationship (≈ precision-bits / 4) follows from the mechanism but a clean in-session re-measurement was not completed; we state the two-parameter structure as **[LOCKED]** by mechanism and mark the precise yield curve as **[OPEN]**.

4.3 Why base sixteen [LOCKED]

The base is how many times the arc is folded per read. A circle bisects cleanly — the perpendicular bisector, the only clean division — so repeated bisection gives 2, 4, 8, 16: powers of two. Base 16 is four bisections per digit. This is not cosmetic: the digit-extraction shift 16^n mod (8k+j) is cheap precisely because $16 = 2^4$ aligns with the binary period structure of the denominators. A base of ten (= 2 × 5) breaks that alignment — the factor of five has no clean place on the bisected arc — which is exactly why there is no base-ten BBP spigot and why decimal π reads as noise. Base 16 reads π along its own fold lines.

5. The Seam: Floating Versus Pinned

The one structural difference between the two machines is the location of the **seam** — the boundary where linear integer accumulation gives way to non-linear extraction.

	BBP (read)	**SHA-256 (write)**
Carrier	π (fixed, public)	schedule field + round constants K
Address	digit position n	the input message
Seam	floats to k = n	pinned at the expansion boundary
Residue	hex digit window	256-bit digest
Invertible?	in principle — seam tracks the query	blocked from a still — seam is fixed

BBP says: move the seam to the address and read the wave. SHA-256 says: force every message through the same fixed seam, fold and carry and mix, and emit the terminal residue. **BBP filters the carrier; SHA pins the carrier and folds it.** The pinned seam is the source of one-wayness — not because the rounds are irreversible (they are not, given the schedule), but because a single still gives no foothold to walk the fixed seam backward.

6. Where the Hardness Actually Lives

Two prior results in the corpus sharpen the location of SHA-256's security, and both are consistent with the still/stream separation.

6.1 The schedule is transparent given the seam transcript [LOCKED]

Because σ_0 and σ_1 are invertible aperture maps over $GF(2)^{32}$, the schedule recurrence can be run backward *when the relevant words are known*: from W[16…30] together with the W[0] anchor one recovers W[1…15], with full sixteen-word dependency coverage emerging by W[26]. The schedule is not an opaque destroyer; it is a transparent recurrence with a readable seam — provided the seam transcript is available. This is the stream being legible, restated in the schedule's own algebra.

6.2 A single still is under-determined [LOCKED]

The flip side is the still being illegible, and it has an exact algebraic form. A single schedule value W[t] is one equation in four unknown parents (the σ_1, raw, σ_0, and raw legs). One equation cannot fix four unknowns: inversion from a lone W[t] is under-determined. This is the same lossiness measured abstractly in Section 2.1

(0.6% injective for one fold), now appearing as the under-determination of one recurrence row. The stream replays forward exactly; only backward-from-a-single-still is blocked.

6.3 The reframed inverse problem [OPEN]

Putting the two together relocates the preimage problem. The hard step is not digest → message directly; it is digest → compatible seam transcript. Formally, the open question is whether the digest constrains the set of compatible seam transcripts tightly enough to reduce the search below brute force. We state this as the central **[OPEN]** problem and explicitly do **not** claim a preimage method. What is established is the *shape* of the problem: a transparent, invertible aperture chain conditioned on a transcript that a single still does not reveal.

Methodological correction, recorded. We reject framing this crossing as a frontal back-solve. Handing a general SMT solver the round equations and asking it to traverse the pinned seam backward fights the seam directly; in session this saturated quickly (an all-free reduced-round solve was instant at four rounds and then stalled by eight rounds — coinciding with the diffusion cone filling). The instinct that inversion must be read orthogonally to the stream rather than along it is retained as a heuristic, while noting that the orthogonal residue-class axis we tested (seed-exit word modulo 11) came back uniform across 5,000 inputs — itself a result (Section 7).

7. Uniformity Is an Indicator, Not an Absence

When we read the seam-exit word modulo 11 across 5,000 inputs, the distribution was flat. The naive reading is *dead end — no signal*. The correct reading is the opposite. A random process gives a lumpy distribution; a flat distribution across classes is the signature of a **complete, unbiased map** — a perfect sort. The only way to be simultaneously uniform (every class equally occupied), unique (no collisions on the full state), and repeatable (deterministic) is a bijection onto a complete set. Uniformity is therefore the **indicator** that the underlying map is complete, which is precisely what makes the output of SHA-256 a hash: uniform because complete, unique because bijective, repeatable because fixed. The flatness was never the absence of structure; it is the fingerprint of completeness, and it is why the orthogonal axis offers no bias to exploit.

Findability obeys a logarithmic horizon [LOCKED]. A related corpus experiment searched the first 2×10^8 digits of π for power-values and found that findability collapses past roughly 8–9 digits. This is exactly the uniform-random prediction: in N digits, a target longer than log-base(N) is expected to appear less than once, and $\log_{10}(2\times10^8) = 8.3$. The cliff is not a defect in π; its landing precisely at the information-theoretic horizon is further evidence of a complete, unbiased field. To locate L digits of content costs about base^L of address space — the same address-versus-content compression that makes the number line behave like an addressable store (a seven-digit index reaching a million digits of content is a ratio of 142,857 : 1).

8. Discussion: The Compiler We Are In

Read together, the results describe one machine and one instruction. The **atom** is the fold — reversible, conserving, order-2. The **instruction** is the stream — the composed recurrence, which alone is injective and reversible and therefore legible. The **grammar** is the dependency ledger — what feeds what, anchored on the root-cover axis. The **still** — the digest, the single digit, the single word — is only the residue: a position, not the operator that produced it.

This reframes computation in the family as **address recognition over stored structure**, with two directions. BBP is the public, solved direction: address plus read-head plus carrier returns stored content. SHA-256 is the locked direction: the carrier is pinned, the read-head (the inverse) is unknown, and recovering it is the open seam-transcript problem. The corpus program — the aperture skeleton, the hinge law, the taxonomy, the root-cover axis, the sigma-inversion laws — is the attempt to reconstruct that read-head from the stream rather than to brute-force it from the still.

Final collapse. **The fold is the atom. The stream is the word. The ledger is the grammar. The still is only the residue.**

9. Status Summary

For clarity, the load-bearing claims and their honest status:

- **[LOCKED]** Single fold against a fixed carrier is lossy: 1,181 / 200,000 distinct (0.6%); residue against fixed C non-injective (921 / 100,000).
- **[LOCKED]** Composed eight-step stream is injective (200,000 / 200,000) and reversible (8/8 recovered); 1-bit input ⟶ 18-bit diffusion.
- **[LOCKED]** SHA-256 schedule values exact (W[16]=61626380, W[17]=000F0000); forward replay from ledger exact (48/48); W[1] root-cover (parents of W[16] and W[17], propagating to W[63]); 1-bit W[1] flip changes 49/64 rows, max 22 bits at t=55.
- **[LOCKED]** Aperture skeleton {0,1,9,14}; W9 Hinge Law and Seed Echo Taxonomy (all 16 seed words classified); Root-Cover Axis ($P16 \cap P17 = \{1\}$). Verified by recurrence with zero violations in the corpus.
- **[LOCKED]** BBP evaluates π to working precision; first hex digits 243F6A88…; two-parameter (location, duration) read-head structure by mechanism; base-16 forced by clean arc-bisection; findability horizon = log-base(N) (cliff at 8.3 for $2{\times}10^8$ digits).
- **[LOCKED]** Schedule transparent given the seam transcript (W[16..30] + W[0] ⇒ W[1..15], full coverage by W[26]); single schedule word under-determined (4 unknowns, 1 equation).
- **[OPEN]** Whether the digest constrains the compatible seam transcript tightly enough to reduce preimage search below brute force. No preimage method is claimed.
- **[OPEN]** Exact BBP digits-per-bit yield curve (clean re-measurement not completed in session).

- **[NULL]** π's hex stream running as structured code on a generic 16-op ISA (within the random band, $z \approx +1.85$; structure was the machine's, not π's). A frontal SMT back-solve of the full schedule (stalls by ~8 rounds at the diffusion-cone boundary).

A-Mark9 / Nexus Recursive Harmonic Framework — Phase 1163+. Every [LOCKED] figure traces to code executed in the session producing this paper; corpus results are cited as verified-by-recurrence with zero violations. Negative results are preserved as navigational data, not discarded.